Adaptive Meshing of Non-Manifold Parametric Patch Complexes with Shared-Curve Compatibility

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Abstract

We present a compatibility-preserving adaptive meshing pipeline for non-manifold parametric patch complexes. In this setting, multiple surface patches may be incident to the same curve, making independent refinement prone to incompatible boundary discretizations, cracks, and unstable transitions near junctions. The proposed method addresses this issue by enforcing shared-curve synchronization as a first-class constraint throughout the adaptive process. Starting from a coarse patch-consistent mesh, the pipeline iteratively evaluates the current discretization, updates refinement factors based on curvature mismatch, and rebuilds the mesh through two coordinated stages. Shared curves are first rediscretized so that all incident patches reuse the same boundary sampling, after which patch interiors are reconstructed in the parametric domain using a combination of 1-irregular quadtree refinement and advancing-front operations. A constrained finalization stage applies smoothing and local cleanup while preserving geometric consistency. The method is evaluated using normalized global error, element-shape quality, and a compatibility metric that measures boundary coherence across shared curves. Across three representative evaluation cases, the approach reduces normalized global surface error by 73.95% to 96.33% while maintaining zero mean compatibility error. In addition, a controlled comparison against internal reference baselines shows that shared-curve synchronization is the key mechanism for preserving compatibility during adaptive meshing in non-manifold configurations, while hybrid reconstruction is the main mechanism responsible for stronger geometric improvement. These results show that the proposed pipeline achieves geometric adaptation while preserving consistent boundary discretization across shared multi-incident curves, supporting robust meshing of planar, curved, and mixed surface regions.

Keywords: adaptive meshing , non-manifold surfaces , patch complexes , shared-curve synchronization , curvature-guided refinement , advancing-front meshing

1 Introduction

Surface mesh generation is a fundamental problem in computer graphics, geometric modeling, and geometry processing because it converts continuous surface descriptions into discrete structures suitable for visualization, simulation, numerical analysis, and downstream geometric computation. Classical solutions include Delaunay-refinement methods with formal guarantees on quality and termination [1–4], as well as remeshing and mesh-improvement techniques designed to enhance sampling quality and adapt local resolution to geometric variation [5].

Many CAD-like and engineered models, however, do not satisfy manifold assumptions. They may contain multi-incident edges shared by three or more surface sheets, Y/T junctions, degenerate contacts, and related singular configurations that challenge meshing pipelines designed for regular manifold connectivity. In such settings, neighboring patches cannot be discretized independently without risk: even when local refinement decisions are reasonable in isolation, they may produce incompatible samplings along shared curves, leading to cracks, unstable transition regions, and poor element quality near topological singularities.

In this work, we consider a piecewise-smooth parametric patch complex in which geometry is represented by surface patches connected through explicit boundary and junction curves. In this representation, compatibility along shared curves is not a secondary implementation detail, but a structural requirement.

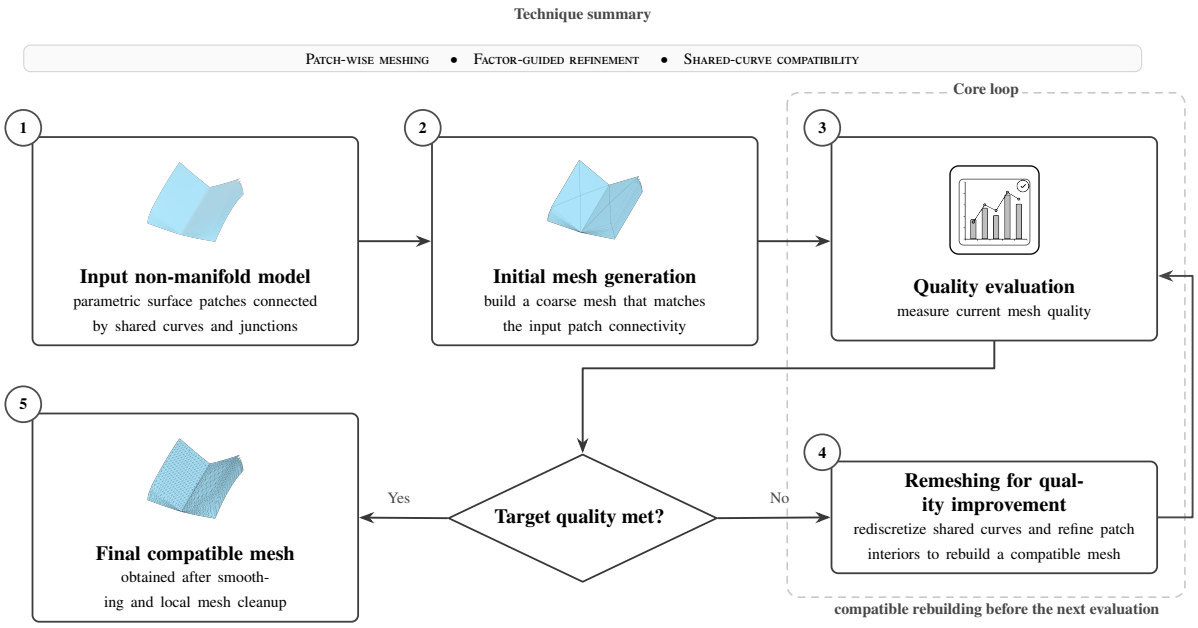


Figure 1: Overview of the proposed compatibility-preserving adaptive meshing pipeline. Starting from a coarse discretization, the method iteratively evaluates geometric error, updates refinement factors, synchronizes shared-curve discretizations across incident patches, and reconstructs patch interiors. The final result achieves improved geometric fidelity while preserving consistent boundary sampling in non-manifold configurations.

Patches incident to the same curve must agree on its discretization so that the resulting mesh remains geometrically and topologically coherent, especially at multi-incident junctions where local inconsistencies can propagate to several neighboring patches at once.

Prior work addresses important parts of this broader problem from different perspectives. Volumetric and grid-based methods such as Dual Contouring can reconstruct surfaces while preserving sharp features [6], but they do not directly provide a curvature-guided adaptive meshing pipeline over parametric patch complexes with explicit shared-curve compatibility. Restricted Delaunay and Frontal-Delaunay methods for piecewise smooth and piecewise linear complexes have significantly advanced robust feature-aware meshing in the presence of curves and corners [7–9]. Advancing-front and hybrid methods, in turn, remain attractive because they offer strong local control near constrained boundaries and geometrically sensitive regions [10, 11].

Despite these advances, an important gap remains between feature-aware meshing in general and adaptive meshing for non-manifold parametric patch complexes in particular. Existing methods provide strong ingredients for robust meshing, local size control, or feature preservation, but they do not directly target the combination of requirements that is central in our setting: operating on non-manifold parametric patch complexes, enforcing explicit compatibility along shared multi-incident curves, and coupling this compatibility constraint with geometry-guided local refinement. This combination is precisely where practical difficulties arise, because boundary synchronization cannot be treated as a post hoc repair when neighboring patches are rebuilt adaptively and repeatedly.

A second challenge lies in evaluation. On models with substantial planar structure, geometric fidelity and isotropic element-shape quality may evolve in a largely compatible way. On more curved and locally anisotropic surfaces, however, these two aspects need not coincide. In such cases, a mesh may substantially improve the geometric approximation of the target surface while still containing triangles that are less isotropic in 3D. This distinction is important because our primary objective is adaptive geometric fidelity under compatibility constraints rather than isotropic element regularity in isolation.

To address this gap, we propose a hybrid adaptive meshing method organized as an iterative evaluate-

and-rebuild loop. Starting from a coarse mesh consistent with the input patch connectivity, the method evaluates the current discretization, updates refinement factors from curvature mismatch, and rebuilds the mesh through two coordinated stages. First, shared curves are rediscretized so that all incident patches reuse the same boundary sampling. Second, patch interiors are reconstructed in the parametric domain through a hybrid strategy that combines structured refinement with local front-based operations. After the target quality is reached, a constrained finalization stage applies smoothing and local cleanup while preserving geometric consistency.

The central hypothesis of this work is that shared-curve compatibility must be enforced as a first-class constraint during adaptive meshing, rather than treated as a post-processing concern. To this end, we design a compatibility-preserving pipeline in which all patches incident to a shared curve reuse the same parameter sampling throughout the adaptive process.

This design eliminates boundary inconsistencies by construction and allows local geometric refinement to proceed without compromising the coherence of the underlying non-manifold patch complex. In contrast to approaches that enforce compatibility after independent refinement, our method guarantees consistent boundary discretization before interior reconstruction.

The proposed approach is evaluated experimentally through geometric error reduction, a dedicated compatibility metric, and a controlled comparison against internal reference baselines. The reported evidence indicates that shared-curve synchronization is the central mechanism for preserving boundary coherence during adaptive meshing in non-manifold configurations, while stronger geometric improvement depends more directly on the hybrid reconstruction stage. The method is designed for non-manifold models that may contain planar, curved, and mixed surface regions, and the experimental design is intended to reflect this geometric diversity without overstating the present comparative scope.

Contributions The main contributions of this work are:

- A compatibility-preserving adaptive meshing pipeline for non-manifold parametric patch complexes, designed to operate directly on models with shared multi-incident curves and junctions;
- A shared-curve rediscretization mechanism that forces all incident patches to reuse a common boundary sampling, reducing mismatch and instability near non-manifold junctions;
- A hybrid patch-interior reconstruction strategy that combines structured refinement and advancing-front operations while preserving boundary compatibility;
- An iterative quality-driven workflow that rebuilds the discretization from a coarse compatible mesh and evaluates it primarily through geometric fidelity, with shape-quality indicators used as complementary descriptors.

Figure 1 summarizes the proposed pipeline and the interaction between evaluation, compatible rebuilding, and finalization. The remainder of the paper is organized as follows. section 2 reviews related work, section 3 presents the proposed method, section 4 reports the experimental results, section 5 discusses limitations and future work, and section 6 concludes the paper.

2 Related Work

Adaptive meshing for parametric surface models lies at the intersection of feature-aware meshing on PSC/PLC domains, hybrid structured–unstructured surface meshing, geometry-driven adaptation, and robust reconstruction for difficult CAD inputs. Since our goal is to generate compatible adaptive meshes on non-manifold parametric patch complexes, the most relevant prior work is the literature that explicitly handles curves, corners, local sizing control, parameterization-based remeshing, and boundary-sensitive reconstruction. A focused comparison is given in Table 1.

Table 1: Focused comparison with the closest methodological families. "Preserves non-manifold structure" indicates whether the method is designed to operate without first repairing or manifoldizing the input. "Explicit shared-curve synchronization" indicates whether the method enforces a common discretization along shared lower-dimensional features during adaptation or reconstruction.

Family	Reference(s)	Adaptive	Preserves non-manifold structure	Explicit shared-curve sync.	Formal guarantees	Anisotropy	CAD robustness	Positioning / limitations relative to this work
Delaunay refinement	[1, 3, 4]	Partial	No	No	Yes	No	Partial	Strong theoretical support for quality-driven meshing, but not designed to preserve a non-manifold parametric patch-complex structure with synchronized shared-curve discretization
PSC/PLC feature-aware meshing	[7–9, 12, 13]	Partial	Partial	Partial	Yes/partial	No	Partial	Closest line in terms of explicit lower-dimensional feature treatment; however, compatibility across shared multi-incident curves in adaptive non-manifold parametric patch complexes is not the central target
Hybrid structured + front-based meshing	[10, 14–16]	Yes/partial	No	No	No/partial	No	Partial/strong	Strong local control near constrained boundaries and difficult CAD regions, but typically without an explicit mechanism for synchronizing shared-curve sampling across all incident patches
Curvature / metric-driven adaptation	[5, 17–19]	Yes	No/partial	No	Partial	Yes/partial	Partial	Provides geometry-aware sizing and, in some cases, anisotropic adaptation or optimized size-field evolution, but usually assumes a smoother connectivity setting and does not explicitly enforce synchronization on shared non-manifold curves
Robust meshing for defective CAD inputs	[16, 20–22]	Partial	Partial	No	No/partial	No	Yes	Strong robustness for dirty CAD geometry, discrete reparametrization, and conforming reconstruction, but focused more on defective input repair/reconstruction than on compatibility-preserving adaptive meshing over a retained non-manifold patch structure
Topology repair / manifoldization	[23, 24]	No	No	No	No	No	Yes/partial	Highly relevant for problematic geometry in practice, but solves a different problem by regularizing or repairing the input rather than preserving the original non-manifold patch-complex connectivity during adaptation
This work	–	Yes	Yes	Yes	No formal proof	No	Partial	Compatibility-preserving adaptive rebuilding for non-manifold parametric patch complexes using shared-curve rediscretization and hybrid patch-interior reconstruction

2.1 Feature-aware meshing on PSC/PLC and CAD-like models

Classical Delaunay-refinement methods establish the foundations of quality-guaranteed mesh generation in 2D and 3D under standard assumptions [1, 3, 4, 25]. In the presence of curves, corners, and surface features, restricted Delaunay approaches over piecewise linear and piecewise smooth complexes (PLC/PSC) became a central framework for feature-aware meshing, supported by both theoretical analysis and practical implementations [7, 8, 12, 13]. Closer to our setting, conforming meshing methods with explicit treatment of junction curves further highlight the importance of preserving consistency along lower-dimensional features [9].

Among the prior lines reviewed in this paper, PSC/PLC feature-aware meshing is the closest conceptual starting point for our work because it keeps curves and corners explicit and recognizes that lower-dimensional features must constrain the final discretization. However, the setting considered here is more specific: we assume a non-manifold parametric patch complex whose shared-curve structure must be preserved during repeated adaptive rebuilding, rather than a pipeline in which feature treatment is present but cross-patch synchronization is not itself the central adaptive constraint.

Recent CAD-oriented work further reinforces the importance of parameterization-aware remeshing and conforming discrete representations. Beaufort et al. [20] show how robust reparametrization pipelines can support remeshing of discrete surfaces, while Yang et al. [22] construct a conforming discrete representation for industrial CAD models before remeshing. These works are highly relevant to our context, but they target conforming reconstruction and remeshing support more broadly, rather than explicit compatibility-preserving adaptive rebuilding on non-manifold patch complexes.

2.2 Hybrid meshing with structured background and advancing front

Hierarchical structures such as quadtrees and octrees are widely used to encode local sizing information and to organize level transitions through finite 1-irregular templates, making them effective as background scaffolds for adaptive meshing [11]. Advancing-front methods, in turn, remain attractive because they naturally preserve boundary information and provide strong local control during element creation [10, 14, 15].

Recent advancing-front work has also emphasized robustness on difficult CAD data. Liu et al. [16] propose an advancing-front framework for dirty composite surfaces, using a parametric-plane construction and robust front checking to handle CAD errors such as gaps and overlaps. This line is especially relevant because it shows how front-based reconstruction can remain effective under geometric difficulty. Our method follows this hybrid spirit, but targets a different bottleneck: not only robust local filling, but also the need to keep a common discretization along shared multi-incident curves while rebuilding neighboring patch interiors consistently.

2.3 Curvature-driven adaptation and constrained mesh improvement

Surface remeshing and mesh optimization methods in geometry processing typically focus on improving local quality, regularity, and feature preservation through local operators and constrained smoothing [5, 26]. In parallel, metric-based adaptation formalizes the relation between geometric or numerical error and element sizing, including isotropic and anisotropic variants [17, 18]. Our work is closer to this line than to pure remeshing: the mesh is not only improved locally, but rebuilt from updated size targets derived from geometric behavior.

Recent work has also refined how size fields and parameterization-domain remeshing can be coupled in CAD contexts. Wang et al. [19] propose alternating size-field optimization and remeshing in the parameterization domain, illustrating how geometry-aware sizing, smoothing, and remeshing can be integrated in a modern CAD pipeline. At the same time, an important distinction must be made. Standard metric-adaptation pipelines primarily focus on how to encode and propagate a sizing field, whereas our central concern is how to couple adaptive refinement with compatibility preservation on shared non-manifold curves. In other words, the main challenge in our setting is not only where to refine, but how to ensure that the refinement decisions made around a shared curve remain coherent across all incident patches.

2.4 Robust meshing for defective and non-manifold inputs

A separate line of work emphasizes robustness on noisy, incomplete, or topologically inconsistent inputs. Methods such as TetWild and ManifoldPlus aim to recover valid meshes from problematic geometry, often by repairing or regularizing the topology before the final discretization [23, 24]. These methods are highly relevant in practice, but their objective differs from ours: instead of repairing the model into a manifold representation, we assume a non-manifold parametric patch complex as input and seek to preserve its shared-curve structure during adaptive meshing.

For CAD-oriented inputs, recent work has pushed robustness further on dirty geometry and conforming reconstruction. Zhang et al. [21] focus on automatic surface mesh generation for CAD models with dirty geometry, while Yang et al. [22] construct a conforming discrete representation that supports later parameterization-based remeshing. These works are close in application domain, but they still address a different core problem from ours: improving robustness and conformity of complex CAD inputs rather than preserving an explicitly non-manifold shared-curve organization throughout adaptive rebuilding.

This difference is not merely terminological. Repair-based or conformity-first pipelines can be entirely appropriate when the goal is to obtain a clean manifold-compatible surface for downstream processing. By contrast, in the present work the non-manifold connectivity is part of the input representation that must remain meaningful throughout the adaptive process. For this reason, preserving the original shared-curve organization is not optional, but part of the problem definition itself.

2.5 What is still missing and why it matters

Taken together, the literature above provides strong ingredients for robust meshing, feature preservation, local size control, and reconstruction under difficult geometry. However, none of these lines directly targets the full combination of requirements that defines our setting: operating on non-manifold parametric patch complexes, preserving the original shared-curve structure, enforcing a common discretization along shared multi-incident curves, and coupling this compatibility constraint with curvature-guided local refinement.

This missing combination matters for a practical reason. In adaptive non-manifold meshing, compatibility cannot be treated as a purely local or post hoc correction. If neighboring patches are refined independently and only later reconciled, the resulting boundary samplings may already be inconsistent, especially around multi-incident curves and junction fans. In such cases, the cost of restoring coherence is not limited to a local edge fix, because the inconsistency can propagate to several adjacent patch interiors and destabilize the quality of the rebuilt mesh.

The contribution addressed in this paper therefore does not lie in proposing adaptive meshing, feature-aware meshing, or front-based reconstruction in isolation. Rather, it lies in the explicit integration of these ingredients under a preserved non-manifold patch-complex representation, where shared-curve compatibility must remain a structural constraint throughout the adaptive process. This is the specific methodological gap that motivates the present work.

The closest prior art to our contribution is not a single method, but a set of adjacent lines that solve different parts of the problem. Liu et al. [16] show that advancing-front reconstruction can be made robust on dirty composite surfaces, but their emphasis is on local front robustness under CAD defects rather than on enforcing a common discretization along shared multi-incident curves during repeated adaptive rebuilding. Wang et al. [19] couple size-field optimization with parameterization-domain remeshing, but their main contribution is to improve how refinement is distributed, not how compatibility is preserved when several patches must reuse the same boundary sampling. Zhang et al. [21] and Yang et al. [22] push robustness and conformity further for industrial CAD models, yet they primarily target dirty-geometry handling and conforming discrete reconstruction before remeshing, rather than preserving an explicitly non-manifold shared-curve organization throughout the adaptive process. Our claim is therefore more specific: the missing piece is the explicit integration of adaptive rebuilding, shared-curve synchronization, and patch-wise reconstruction under a preserved non-manifold parametric patch-complex representation.

2.6 Positioning of this work

The proposed method combines ideas from feature-aware PSC meshing, hierarchical background discretization, advancing-front reconstruction, and curvature-driven adaptation, but its distinguishing aspect is not any one of these ingredients taken in isolation. The central difference is that compatibility along shared multi-incident curves is treated as a first-class adaptive constraint. In practical terms, all patches incident to the same curve are forced to reuse the same updated boundary sampling before interior reconstruction begins. More specifically, unlike prior work that either emphasizes feature-aware meshing without explicit adaptive synchronization across all incident patches, or robust/front-based reconstruction without shared-curve compatibility as the central adaptive mechanism, our method rebuilds the discretization from a coarse compatible mesh while forcing all incident patches to reuse the same updated curve sampling before interior reconstruction. Under this interpretation, the method is best understood as a compatibility-preserving adaptive rebuilding strategy for non-manifold parametric patch complexes, rather than as a generic remesher, a dirty-CAD repair pipeline, or a purely metric-adaptation framework.

To the best of our knowledge, prior work provides important and closely related ingredients, but does not directly target this particular combination under a preserved non-manifold parametric patch-complex representation. The contribution of this paper lies in making this combination explicit and operational: adaptive rebuilding, shared-curve synchronization, and patch-wise reconstruction are treated as mutually dependent parts of a single compatibility-preserving workflow.

This positioning is important for interpreting the scope of the paper correctly. The manuscript does not claim a universal surface-meshing solution, nor formal superiority over all alternative meshing strategies. Instead, it contributes a methodologically specific pipeline for a practically relevant setting: adaptive meshing on non-manifold parametric patch complexes in which shared-curve coherence must be preserved throughout the rebuilding process. Under this interpretation, the main value of the work is not generic remeshing performance alone, but the explicit integration of compatibility-preserving constraints with geometry-driven adaptation.

Algorithm 1 Implementation-aligned adaptive meshing pipeline

Require: Parametric patches $\{\Phi_k : U_k \to \mathbb{R}^3\}$, explicit shared curves Γ, maximum iterations $N_{\max}$, tolerance ϵ, stagnation threshold τ_E, patience P

Ensure: A compatible mesh across all shared curves, including multi-incident ones

```
Read the patch complex and build the patch–curve incidence structure
Generate a coarse initial mesh on each patch
Evaluate the initial global mismatch E^0
for t = 1 to N_max do
    Compute analytic/discrete curvature on the current mesh and evaluate E^t
    if E^t ≤ ε then
        break
    end if
    Update patch- and curve-level subdivision factors
    Rediscretize all shared curves and update their parameter samples
    Rebuild each patch interior in parameter space
    Apply restricted Laplacian smoothing on interior vertices
    if relative improvement remains below τ_E for P consecutive iterations then
        break
    end if
end for
Export the final mesh and per-iteration quality/error logs
```

3 Proposed Method

We propose an adaptive meshing pipeline for non-manifold piecewise-smooth parametric patch complexes with explicit shared curves. The method follows an iterative *evaluate–adapt–rebuild* loop, but differs from classical metric-field-based adaptive remeshing in three practical aspects. First, refinement is *factor-driven*: the implementation updates global, patch-level, and curve-level subdivision factors from curvature mismatch and local geometric complexity, rather than evolving a continuous background metric field $h(u, v)$ [17, 18, 25]. Second, meshing is performed primarily in the *parametric domain* of each patch, using the surface map only to evaluate geometry and to place the final surface vertices. Third, compatibility across boundary, junction, and multi-incident curves is enforced by reusing the same shared curve object and the same updated parameter sequence on all incident patches.

The pipeline is designed for non-manifold models that may contain planar, mildly curved, and strongly curved regions within the same patch complex. For this reason, the method is driven primarily by a geometric fidelity signal derived from curvature mismatch rather than by a purely isotropic notion of triangle regularity. Shape-quality measures remain useful for monitoring and local control, but they are not the principal driver of the adaptive process.

At a high level, each iteration performs the following steps: (i) evaluate the curvature mismatch between the current mesh and the analytic patch geometry; (ii) update patch- and curve-level refinement factors; (iii) rediscretize shared feature curves; (iv) rebuild patch interiors in parameter space using a structural quadtree, optional templates, and a feature-aware advancing front; and (v) improve the resulting mesh by restricted smoothing in parameter space. The current implementation supports parallel infrastructure in the code base, but the workflow described in this paper corresponds to the sequential experimental mode used in our evaluation.

3.1 Input Model and Data Structures

The input is a piecewise-smooth parametric patch complex represented as a set of patch maps $\{\Phi_k : U_k = [0, 1]^2 \to \mathbb{R}^3\}$ connected by explicit feature curves. Each curve stores the list of incident patches, and we classify a curve as *non-manifold* whenever it is incident on three or more patches. This representation

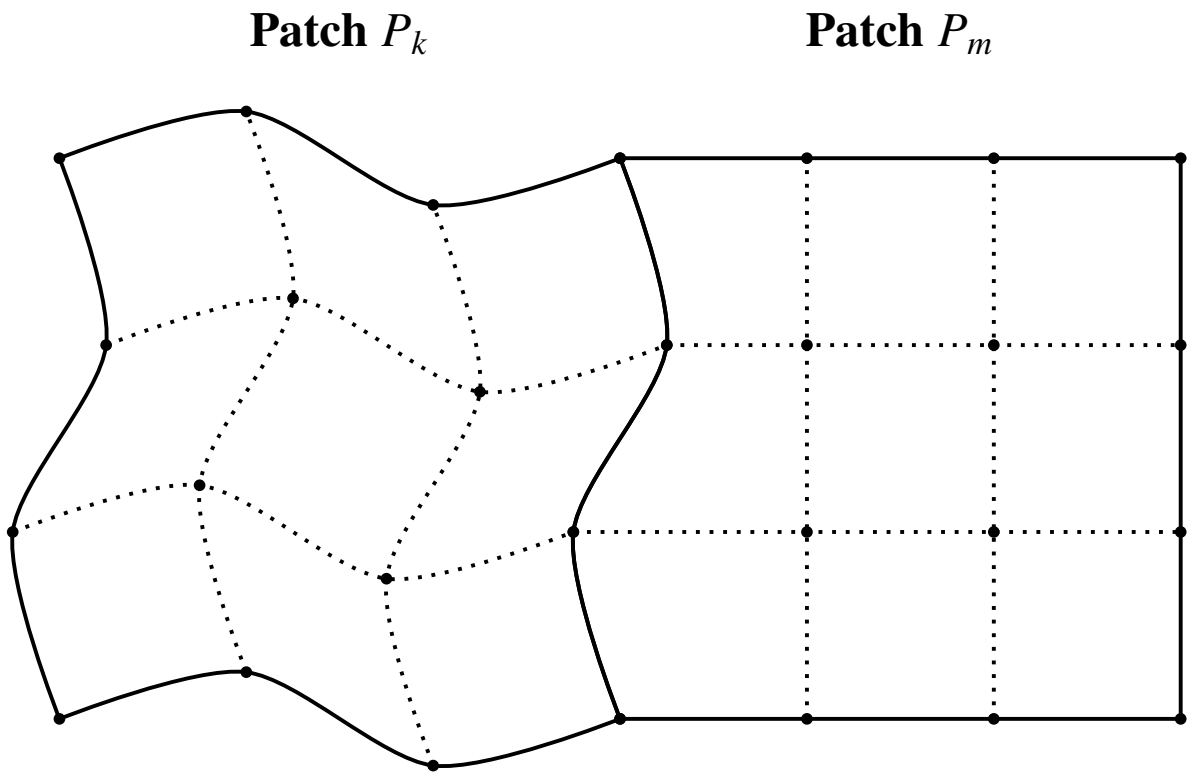


Figure 2: Non-manifold patch adjacency with shared curves. Multiple surface patches are incident to the same boundary curve, requiring consistent discretization across all incident domains. Independent refinement of each patch leads to incompatible boundary sampling and visible mismatches, motivating the need for explicit shared-curve synchronization.

is consistent with feature-aware meshing formulations that keep curves and corners explicit rather than reconstructing them a posteriori [7, 9, 13].

The two core topological objects in the implementation are therefore:

1. a **patch object**, which stores its parametric map and its local submesh; and
2. a **curve object**, which stores both its current point list and the set of incident patches.

Because a curve object is shared by all incident patches, any update of its point list or of its parameter samples is immediately visible to every patch that uses it. This shared ownership is the basis of the compatibility policy used later in the pipeline.

Each patch also stores a *submesh* in the parametric domain. Vertices keep both spatial coordinates and the corresponding parametric information needed to re-evaluate curvature and remap the mesh through Φ_k. The current implementation starts from a deliberately simple coarse mesh: the four corners of the unit square and its center are sampled, and four initial triangles are created from this five-point stencil. This initialization is independent of the quadtree/templates stage and provides a stable starting mesh for the first error evaluation.

3.2 Connectivity Bookkeeping and Shared-Curve Policy

The implementation does not rely on a heavy geometric preprocessing stage. Instead, it performs lightweight topological bookkeeping directly on the patch and curve data structures. Before each adaptation step, the points of each parametric curve are reordered according to their parameter value, and the corresponding parameter list is updated. Since curves are shared objects, this ordering becomes the common ordering seen by all incident patches.

This behavior is stronger than a mere local boundary synchronization. When a curve is incident on multiple patches, the same updated parameter sequence is reused by all of them. As a consequence, the method maintains identical vertex ordering along the shared curve on every incident patch, which prevents cracks and inconsistent boundary triangulations. In practice, the same mechanism is used for both manifold and non-manifold curves; for multi-incident curves, the common sequence acts as the compatibility anchor for the whole fan of adjacent patches.

Algorithm 2 Shared-curve ordering and parameter synchronization

1: **for all** feature curves $\Gamma \in \mathcal{C}$ **do**
2: Gather the incident patches $\mathcal{N}(\Gamma)$
3: Sort the current curve points according to the curve parameter
4: Update the shared parameter list of Γ
5: Reuse this same parameter list on all patches in $\mathcal{N}(\Gamma)$
6: **end for**

Compatibility Invariant After each shared-curve update, all incident patches reuse the same ordered parameter sequence:

$$\Gamma = \{u_1, u_2, \ldots, u_n\}. \tag{1}$$

This guarantees that boundary discretizations are identical across all incident patches before interior reconstruction, eliminating mismatch by construction.

3.3 Compatibility Metric

To quantitatively evaluate boundary consistency across incident patches, we define a compatibility error along shared curves.

Let Γ be a shared curve and let $\mathcal{N}(\Gamma)$ denote the set of incident patches. For any pair of incident patches (p, q) sharing Γ, let $x_i^{(p)}$ and $x_i^{(q)}$ denote corresponding boundary points induced by the same synchronized parameter samples along the curve. The pairwise compatibility error is defined as

$$e_{\text{compat}}^{(p,q)}(\Gamma) = \frac{1}{N} \sum_{i=1}^{N} \left\| x_i^{(p)} - x_i^{(q)} \right\|. \tag{2}$$

For synchronized runs, the corresponding points are induced directly by the shared parameter samples. When evaluating unsynchronized baselines in which the incident boundary arrays differ in size, corresponding points are established through closest-point projection along the compared boundary traces. This convention ensures that the compatibility metric remains well-defined even when explicit one-to-one boundary correspondence is lost.

When Γ is incident to more than two patches, the compatibility error of the curve is obtained by averaging over all incident patch pairs:

$$E_{\text{compat}}(\Gamma) = \frac{2}{m(m-1)} \sum_{\substack{p,q \in \mathcal{N}(\Gamma) \\ p<q}} e_{\text{compat}}^{(p,q)}(\Gamma), \qquad m = |\mathcal{N}(\Gamma)|. \tag{3}$$

Finally, the global compatibility indicator reported in the experiments is the mean compatibility error over the set of shared curves $\mathcal{C}_s$:

$$\bar{E}_{\text{compat}} = \frac{1}{|\mathcal{C}_s|} \sum_{\Gamma \in \mathcal{C}_s} E_{\text{compat}}(\Gamma). \tag{4}$$

A value of $\bar{E}_{\text{compat}} = 0$ indicates perfect alignment of boundary discretizations across all incident patches on all shared curves.

This metric directly captures the consistency of shared-curve discretization and complements the geometric error measures used in adaptive refinement.

3.4 Curvature Mismatch and Refinement Factors

Adaptation is driven by the disagreement between the analytic curvature of the patch geometry and the discrete curvature measured on the current mesh. Analytic curvature is computed directly from the patch parameterization, while discrete curvature is evaluated at mesh vertices using standard discrete operators for Gaussian and mean curvature [5, 27, 28].

The implementation uses a simple but important rule when selecting the local curvature signal. At a mesh vertex j, it first tests the magnitude of the analytic Gaussian curvature. If Gaussian curvature is sufficiently large, the comparison is performed with Gaussian curvature; otherwise the code falls back to mean curvature. Denoting the selected quantities by $\kappa_a(j)$ and $\kappa_d(j)$, this choice can be summarized as

$$\kappa_a(j), \kappa_d(j) = \begin{cases} K_a(j), K_d(j), & \text{if } |K_a(j)| \geq \tau_K, \\ H_a(j), H_d(j), & \text{otherwise,} \end{cases} \tag{5}$$

where τ_K is the numerical threshold used by the implementation. This selection mirrors the logic used in the code and avoids basing the adaptation signal on nearly vanishing Gaussian curvature in almost flat regions.

For each patch submesh k, the code accumulates a normalized quadratic mismatch over the vertices of that patch. Let V_k be the vertex set of patch k. The implementation-level patch error is well approximated by

$$E_k = \frac{1}{|V_k|} \sqrt{\frac{\sum_{j \in V_k} \big(\kappa_d(j) - \kappa_a(j)\big)^2}{\sum_{j \in V_k} \kappa_a(j)^2 + \varepsilon_E}}, \tag{6}$$

where ε_E is a small regularizer. The global mismatch is then the mean patch error

$$E = \frac{1}{|\mathcal{P}|} \sum_{k \in \mathcal{P}} E_k. \tag{7}$$

This is the primary quantity used to monitor convergence and to decide whether a new adaptation step is necessary.

In addition to the error signal above, the implementation also uses local curvature magnitude to scale subdivision intensity. We summarize the patch-wise mismatch and curvature indicators through normalized ratios

$$r_E(k) = \frac{E_k}{\max_m E_m + \varepsilon_E}, \qquad r_\kappa(k) = \frac{\kappa_k}{\max_m \kappa_m + \varepsilon_\kappa}, \tag{8}$$

where κ_k denotes a patch-level summary of analytic curvature magnitude, and $\varepsilon_E, \varepsilon_\kappa > 0$ are small stabilizers. These ratios are then converted into bounded subdivision factors. For a shared curve Γ, the implementation uses a conservative factor based on the strongest signal among its incident patches,

$$\begin{aligned} r(\Gamma) &= \max_{k \in \mathcal{N}(\Gamma)} \max\{r_E(k),\, r_\kappa(k)\}, \\ f_\Gamma &= \mathrm{clip}\Big(f_{\mathrm{glob}}\big(1 + \alpha_E\, r(\Gamma)\big),\, f_{\min},\, f_{\max}\Big), \end{aligned} \tag{9}$$

where f_{glob} is the global factor updated from the iteration-to-iteration improvement behavior. A similar bounded rule is used to update patch-level factors for the interior remeshing stage.

The current refinement policy is therefore geometry-driven and essentially isotropic. It reacts to geometric mismatch and curvature magnitude, but it does not yet encode a full anisotropic tensor metric aligned with principal directions. This choice keeps the method operationally simple and tightly coupled to the compatibility-preserving rebuilding strategy developed in this paper.

3.5 Feature-Curve Rediscretization with Shared Sampling

Curve adaptation is performed before the patch interiors are rebuilt. This is crucial: once the curve samples are updated, they become the boundary anchors for the next patch remeshing step. Let h_{bnd} denote a

reference boundary spacing. The target spacing induced by the current curve factor is

$$h_\Gamma = \frac{h_{\mathrm{bnd}}}{f_\Gamma}. \tag{10}$$

In the implementation, curve adaptation is applied in two complementary passes. In the first pass, the code evaluates the intrinsic curve response and updates the parameter subdivision using a binary tree restricted to the curve parameter. In the second pass, the same curve is adapted again using the surrounding surface curvature as the main signal. For curves shared by several patches, this second pass uses a conservative surface signal: midpoint curvature is evaluated on the incident patches and the strongest refinement request is retained. The resulting binary tree is then restricted so that the parameter sequence remains graded.

Because the curve object is shared, the updated parameter list is stored only once and then reused by all incident patches. This is the mechanism that keeps boundary triangulations compatible even when a curve is shared by three or more patches.

Algorithm 3 Shared feature-curve rediscretization

for all feature curves Γ **do**
    Sort the current points of Γ by parameter value
    Build a binary segment tree over the current parameter intervals
    Refine segments whose 3D length is incompatible with h_Γ
    Refine again where the surrounding surface curvature requests a denser sampling
    Apply a restriction operator to avoid abrupt adjacent level jumps
    Extract the new ordered parameter samples and update Γ
    Reuse the same updated parameter list on all patches in $\mathcal{N}(\Gamma)$
end for

3.6 Structural Quadtree and Graded Transitions in the Parametric Domain

For each patch P_k, the method builds or reuses a structural quadtree on the parametric domain $U_k = [0,1]^2$ to (i) anchor the subdivision to the feature polylines induced by the shared curve samples and (ii) enforce graded neighbor transitions. This graded structure prevents abrupt resolution jumps and provides a stable support for subsequent front-based filling [11].

We refine cells intersected by feature segments until a level compatible with local sampling,

$$\ell^\star(s) = \min\Big\{\ell \in \mathbb{N} :\ \mathrm{diag}(Q_\ell) \le \kappa_{\mathrm{seg}}\|\mathbf{b}-\mathbf{a}\|\Big\}, \tag{11}$$

and cap the depth to avoid over-refinement,

$$\ell_{\max}(P_k) = \left\lceil \log_2\!\left(\frac{\mathrm{diag}(U_k)}{\kappa_{\mathrm{bnd}}\, h_{\min}(P_k)}\right)\right\rceil. \tag{12}$$

Whenever possible, the implementation reuses the submesh from the previous iteration and only updates the target sizes. If the reuse fails or becomes inconsistent with the new boundary sampling, the quadtree is rebuilt from scratch.

3.7 Filling via Feature-Aware Advancing Front

After the structural subdivision is available, the interior of each patch is completed with a feature-aware advancing-front (AF) procedure in parameter space. Regular leaves can be filled by structured patterns or templates, whereas transition regions and neighborhoods of shared curves are handled by front propagation. Candidate insertions are constrained by both the local size induced by the quadtree and the shared feature samples. In the current implementation, feasibility requires that the proposed insertion does not introduce

Algorithm 4 Structural quadtree construction with graded balancing

Require: Feature polylines in U_k induced by the shared samples

Initialize the quadtree with the root cell Q_0
for all feature segments $s = [\mathbf{a}, \mathbf{b}]$ in U_k **do**
    Mark cells intersecting s
    Refine marked cells until level $\ell^\star(s)$ or cap $\ell_{\max}(P_k)$
end for
while there exists a leaf adjacency violating graded-transition constraints **do**
    Refine the coarser leaf and update adjacency
end while
Optionally generate template-based interior elements

local intersections, does not invert the front locally, and remains compatible with the local target size. This keeps the final triangulation compatible with the already synchronized curve boundaries and is consistent with classical paving/front strategies adapted to surface meshing [10, 14, 25].

When a preferred insertion is not feasible, the implementation applies local fallback rules intended to preserve closure and forward progress without violating the shared-boundary constraints. These rules are conservative: their purpose is not to optimize local element shape aggressively, but to avoid invalid or unstable completions near constrained regions and recent shared-curve updates.

Algorithm 5 Feature-aware advancing-front filling

Initialize the front with boundary edges and feature polylines from the shared samples
while the front is not empty **do**
    Select a front edge according to local size/quality priorities
    Propose candidate insertions in U_k using the local front geometry and the quadtree cell size
    if validity, local size compatibility, and front-conflict checks pass **then**
        Insert element(s) and update the front
    else
        Apply local fallback rules to preserve closure and progress
    end if
end while

3.8 Surface Mapping and Parametric Storage

The method operates primarily in the parametric domains. Each vertex stores its owning patch identifier k_i and parametric coordinates $\boldsymbol{\xi}_i = (u_i, v_i)$. Whenever geometric evaluation or export is required, the point is mapped to 3D by

$$\mathbf{x}_i = \Phi_{k_i}(u_i, v_i). \tag{13}$$

This design keeps the meshing and smoothing operations stable in U_k while preserving a direct link with the surface geometry.

3.9 Multi-Incident Compatibility Policy

For any curve Γ with $|\mathcal{N}(\Gamma)| \geq 3$, the method enforces three practical rules: (i) all incident patches reuse the same ordered curve samples, (ii) the curve orientation is kept consistent through the shared parameter list, and (iii) junction and boundary vertices remain constrained during front propagation and smoothing. The objective is to keep the vertex sequence along Γ identical across all incident patches without a separate manifoldization stage. In practice, the current implementation applies the same shared-sampling mechanism to all curves, including those incident on three or more patches.

3.10 Smoothing and Final Local Improvement

After filling, the implementation applies restricted Laplacian smoothing *in the parametric domain* on interior vertices while keeping feature, boundary, and junction vertices fixed. This is a standard and efficient way to improve element quality while preserving sharp and constrained features [5, 26]. For an interior vertex i, the update can be written as

$$\boldsymbol{\xi}_i^{t+1} = (1-\varphi)\boldsymbol{\xi}_i^t + \varphi \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \boldsymbol{\xi}_j, \qquad 0 < \varphi < 1. \tag{14}$$

Updates that would invert local elements in parameter space, create locally distorted one-rings, or worsen neighborhoods adjacent to constrained boundaries are rejected. In practice, this rejection step prevents smoothing from trading geometric consistency for an unfavorable one-ring configuration near constrained boundaries or transition zones.

Algorithm 6 Restricted parametric smoothing

for $s = 1$ to N_{smooth} **do**
  for all vertices i **do**
    if i is an interior vertex **then**
      Propose a parametric Laplacian update via Eq. (14)
      Reject the update if it introduces inversion, locally distorted one-rings, or degradation near constrained boundaries
    else
      Keep i fixed as a hard feature/boundary/junction constraint
    end if
  end for
end for
Map vertices to $\mathbb{R}^3$ using Eq. (13)

3.11 Convergence Criterion

The iterative process stops when the global mismatch falls below a tolerance or when improvement stagnates over a prescribed patience window. This can be summarized as

$$\begin{aligned} &E^t \le \epsilon \\ &\quad \vee \left(\forall s \in [t-P+1, t] : \frac{E^{s-1} - E^s}{\max(E^{s-1}, \epsilon)} < \tau_E \right). \end{aligned} \tag{15}$$

This criterion reflects the implementation policy used to avoid both excessive iterations and unstable oscillatory refinements.

3.12 Complexity and Export

Curvature evaluation is linear in the number of vertices and faces per patch when standard one-ring operators are used [27]. Quadtree construction has near $O(N \log N)$ behavior for N visited cells, with practically efficient graded balancing [11]. The advancing-front stage is roughly linear in the number of inserted elements in typical front-based implementations, with the dominant constants associated with local feasibility tests and neighborhood maintenance [10, 14]. The patch-wise organization also makes parallel execution possible in principle, although the present paper only reports the sequential evaluation mode. The final mesh can be exported in `.obj`, `.off`, or `.pos` formats together with auxiliary logs used for the experimental analysis.

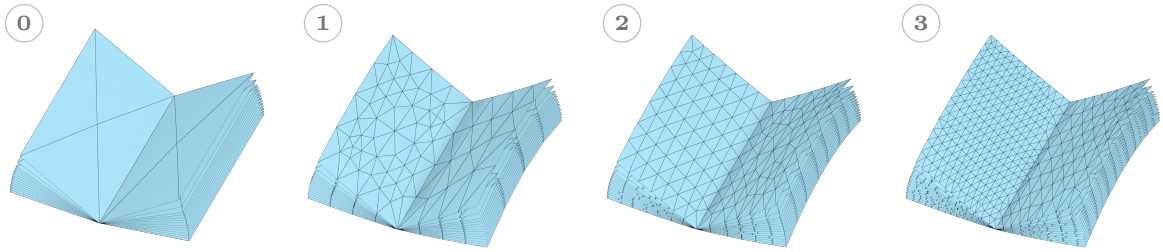


Figure 3: Adaptive meshing sequence for the Book model. From left to right, the mesh evolves from a coarse initial discretization to a refined representation under curvature-driven refinement. Shared-curve synchronization ensures consistent boundary sampling across incident patches at each iteration. The final mesh combines improved geometric fidelity with coherent discretization along multi-incident edges.

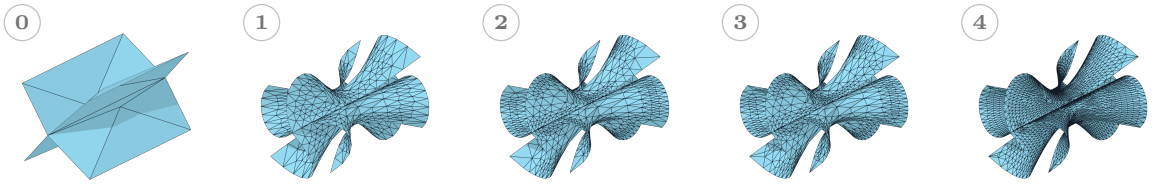


Figure 4: Adaptive meshing sequence for the Eistute model. The mesh is progressively refined under curvature-driven adaptation while preserving consistent boundary sampling across shared curves. Despite smooth curvature variations and anisotropic regions, the method maintains stable refinement and coherent discretization along patch boundaries throughout the process.

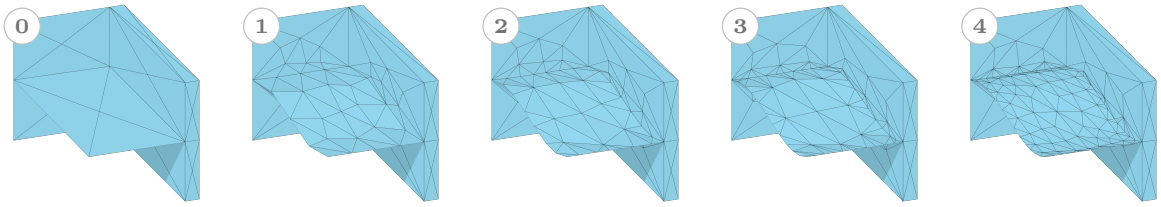


Figure 5: Adaptive meshing sequence for the Decor Shelf model. The method operates on a mixed configuration combining planar and curved patches, progressively refining the mesh while preserving boundary consistency across shared curves. The final result demonstrates robust behavior across heterogeneous geometries, maintaining coherent discretization at interfaces between different surface types.

4 Results

This section evaluates the proposed pipeline on three representative evaluation cases—Book, Eistute, and Decor Shelf—obtained in sequential mode. The goal is not to claim uniformly monotone improvement under all criteria at every stage, nor to argue for broad superiority over alternative meshing backends. Rather, the purpose of the evaluation is to examine, in a focused and verifiable way, two complementary aspects of the method: its behavior on the main non-manifold evaluation cases and its controlled comparison against internal reference baselines designed to isolate the roles of shared-curve synchronization and hybrid

reconstruction within the present pipeline.

Compatibility along shared curves is enforced by construction in the method and inspected qualitatively in the reported mesh sequences. Quantitatively, the present evaluation emphasizes two complementary perspectives. The first is *geometric fidelity*, measured through the global curvature-based surface error used by the adaptive process. This is the principal indicator of whether the rebuilt mesh better approximates the target surface geometry. The second is *element-shape quality*, summarized here through the normalized triangle measure

$$q(T) = \frac{2r}{R}, \tag{16}$$

where r and R are the inradius and circumradius of triangle T, respectively. Values closer to 1 indicate better-shaped and more isotropic triangles.

The normalized global error is reported as

$$E_{\mathrm{norm}}(s) = \frac{E(s)}{E(0)}, \tag{17}$$

so that every execution starts at 1.0000 and later stages are directly comparable across models. In the present section, we use the share of elements with $q \geq 0.60$ as a compact summary indicator of good element shape, while the full quality distribution is visualized through the three exclusive bands $q \leq 0.30$, $0.30 < q \leq 0.60$, and $0.60 < q \leq 1.00$.

This distinction is important for interpretation. On models with large planar or mildly curved regions, such as Book, the isotropic shape indicator $2r/R$ aligns well with visual inspection and with the reduction of geometric error. On more strongly curved and locally anisotropic models, such as Eistute, this same indicator must be read more carefully: a mesh can substantially improve the geometric fidelity of the surface while still containing many triangles that are less isotropic in 3D. For this reason, the current results should be interpreted primarily through the evolution of the global geometric error, with q acting as a complementary indicator of shape regularity rather than as a universal proxy for final surface quality.

Across the three models, Book provides the clearest joint evidence of compatibility-preserving adaptation, strong geometric-error reduction, and favorable element-shape evolution. Eistute is the most informative irregular case, because it shows that strong error reduction on a more curved and anisotropic surface does not necessarily imply equally strong gains in isotropic 3D triangle quality. Decor Shelf occupies an intermediate position, showing substantial improvement in geometric fidelity together with more moderate but still meaningful changes in shape quality. The per-model discussions below preserve the original stage order of the evaluation traces and distinguish intermediate stages from the selected final stage whenever this distinction is important for interpretation.

These three evaluation cases also differ substantially in patch-complex composition. Book contains 28 curved patches, Eistute contains 4 strongly curved patches, and Decor Shelf contains 13 patches, of which only one is curved and the remaining ones are planar. This distinction is relevant not only for geometric interpretation but also for runtime analysis, because the proposed pipeline rebuilds patch interiors individually and therefore accumulates substantially different reconstruction costs across models.

In addition to the geometric and shape-quality measures defined above, we also evaluate boundary consistency across shared curves using the mean compatibility error $\bar{E}_{\mathrm{compat}}$, defined in Section 3.3. This indicator aggregates pairwise compatibility deviations across all incident patch pairs on each shared curve and is then averaged over the set of shared curves considered in the evaluation. This metric complements the normalized global error by explicitly capturing the coherence of discretization across incident patches. The first part of the section reports the behavior of the full method across the three evaluation cases, whereas the later controlled comparison examines how the same metrics change when explicit shared-curve coupling or hybrid reconstruction is removed.

4.1 Book model

The Book model is the clearest and most representative case in the paper. Its qualitative evolution in Figure 3 highlights the shared central curve, the non-manifold patch arrangement, and the favorable interaction between compatibility-preserving rebuilding, geometric-error reduction, and shape-quality improvement.

Table 2: Summary of the selected final stage for each model. The results are reported in terms of normalized global error (Eq. 17), error reduction relative to the initial discretization, and the percentage of elements with shape quality $q \geq 0.60$ (Eq. 16).

Model	Final iteration	Elements	E_{norm}	Error reduction	$q \geq 0.60$
Book	3	8920	0.0843	91.57%	97.04%
Eistute	4	7184	0.0367	96.33%	58.63%
Decor Shelf	4	276	0.2605	73.95%	56.52%

(a) Iterative evolution

Stage	Elements	Norm. error	$q \geq 0.60$
0	112	1.0000	50.00%
1	1424	0.7181	87.99%
2	3334	0.5587	96.28%
3	8920	0.0843	97.04%

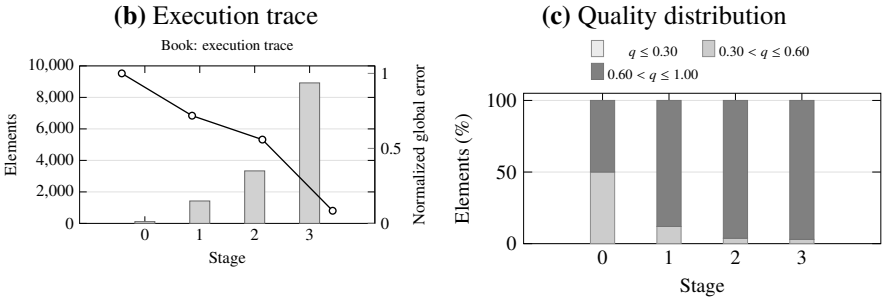


Figure 6: Progress of adaptive meshing on the Book model across iterations. (a) Full sequential trace for the reported evaluation run, reporting the evolution of the number of elements, normalized global error, and the percentage of elements with $q \geq 0.60$ across refinement stages 0–3. (b) Progress plot showing the evolution of element count and normalized global error as the mesh is iteratively rebuilt under curvature-driven refinement and shared-curve synchronization. This model provides the clearest all-around positive case, combining strong geometric improvement with stable adaptive behavior throughout the reconstruction process. (c) Evolution of element-shape quality for the Book model across adaptive iterations. The distribution of triangles across quality groups (Eq. 16) shifts toward higher-quality bins as refinement progresses, indicating improved element regularity without degradation of mesh quality.

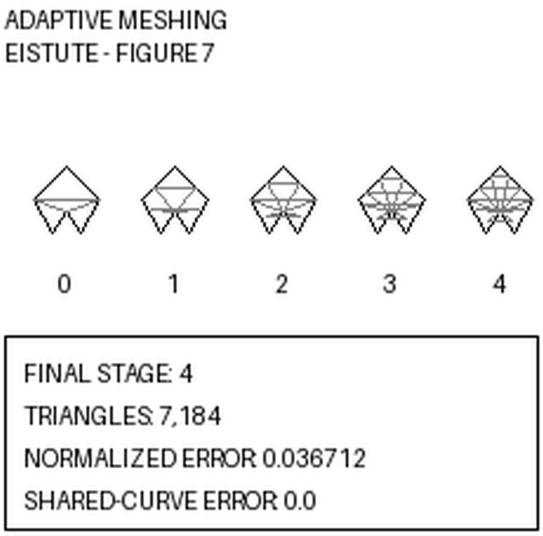
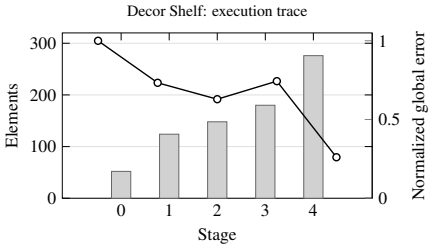


Figure 7: Representative summary of the Eistute adaptive-meshing result used for the GRSI reproduction target. The schematic shows saved stages 0–4. The selected final Stage 4 contains 7,184 triangles, reaches normalized global error 0.036712, and preserves zero mean shared-curve compatibility error.

Book is the strongest overall result among the three evaluation cases because both main evaluation axes evolve in the same favorable direction. As shown in Figure 6(a–b), the normalized global error decreases from 1.0000 in the initial mesh to 0.7181 in Stage 1, 0.5587 in Stage 2, and 0.0843 in Stage 3,

(a) Iterative evolution

Stage	Elements	Norm. error	$q \geq 0.60$
0	52	1.0000	42.31%
1	124	0.7332	55.65%
2	148	0.6290	54.05%
3	180	0.7436	41.11%
4	276	0.2605	56.52%



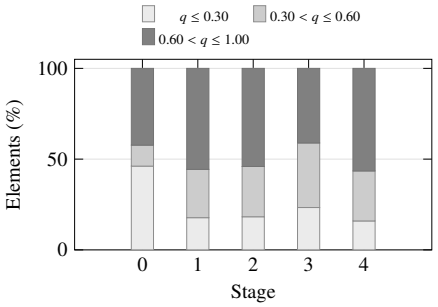


Figure 8: Progress of adaptive meshing on the Decor Shelf model across iterations. The normalized global error evolves non-monotonically, with an intermediate degradation followed by a strong recovery in the final stage, as the mesh is rebuilt under curvature-driven refinement and shared-curve synchronization. Because this model combines planar and curved surface regions within the same non-manifold structure, it provides a representative mixed-geometry case. The observed error reduction indicates that the proposed method remains stable and effective across heterogeneous configurations. (c) Evolution of element-shape quality for the Decor Shelf model across adaptive iterations. The distribution of triangles across quality groups (Eq. 16) remains stable while gradually shifting toward higher-quality bins as refinement proceeds. Because this model combines planar and curved regions within the same non-manifold structure, the plot provides evidence that the proposed method preserves mesh regularity under mixed geometric conditions.

corresponding to a total reduction of 91.57% relative to the initial state. At the same time, the share of elements with $q \geq 0.60$ climbs from 50.00% to 97.04%, indicating that the adaptive rebuilding process not only improves the geometric fidelity of the mesh, but also produces a strongly favorable shape distribution.

The quality distribution in Figure 6(c) reinforces this interpretation. As the execution progresses, the distribution shifts consistently away from the lower and intermediate quality bands and concentrates increasingly in the range $0.60 < q \leq 1.00$. This is the clearest example in the paper of a model for which the isotropic 3D shape indicator aligns well with both visual inspection and geometric-error reduction. In this sense, Book functions as the most complete positive case for the proposed method.

From an interpretive standpoint, Book also clarifies the internal balance of the adaptive pipeline. Stage 2 already provides a very strong quality–error compromise, whereas Stage 3 reaches the lowest reported geometric error among the reported refinement stages. This distinction matters because it shows that the method should not be read as optimizing a single scalar target in isolation; rather, it preserves compatibility while driving refinement toward lower geometric mismatch, with the selected final stage reflecting a practical balance between fidelity and shape quality.

The mean compatibility error $\bar{E}_{\text{compat}}$ remains equal to zero along all shared curves, confirming that boundary discretization is consistently preserved across incident patches.

4.2 Eistute model

Eistute is the most informative irregular case in the paper because it highlights a key methodological point: on more strongly curved and locally anisotropic surfaces, substantial reduction of the geometric error does not necessarily translate into equally strong gains in isotropic 3D triangle quality.

Eistute differs markedly from Book. As summarized in Figure 7, the evaluation run achieves the strongest error reduction of the three evaluation cases, reaching a normalized global error of 0.0367 at the selected final Stage 4, which corresponds to a total reduction of 96.33%. At the same time, however, the compact shape indicator $q \geq 0.60$ remains in a moderate range, rising from 50.00% to 65.41% in Stage 1 and then settling between approximately 58% and 62% in the later stages. If interpreted in isolation, this could suggest only limited quality improvement. In the context of the present model, however, such a reading would be incomplete.

The reported Eistute quality statistics provide a more nuanced picture. The execution does not collapse

into the low-quality band; rather, it redistributes elements between the intermediate and good-quality ranges while aggressively reducing the geometric mismatch to the target surface. Because the underlying adaptation policy is primarily isotropic (as discussed in Section 3.4), preserving strict shared-curve compatibility in highly curved transition regions can force the advancing front to generate locally stretched, and therefore less isotropic, elements in order to close the domain robustly. In other words, when tension arises between geometric fidelity, rigid boundary synchronization, and triangle isotropy, the current method correctly prioritizes compatibility and fidelity over perfectly isotropic element shapes. Eistute is therefore the clearest case in which the surface-curvature error should be regarded as the principal evaluation metric, with isotropic shape quality acting as a complementary rather than dominant criterion.

For the overall interpretation of the method, Eistute is therefore not a contradiction of the main claim, but an important refinement of it. It shows that compatibility-preserving adaptive rebuilding remains effective on more demanding curved geometry, while also illustrating the limitation of relying too heavily on a single isotropic 3D quality indicator. In the present paper, Eistute is the strongest evidence that the method adapts not only across non-manifold topology, but also across more challenging mixtures of curvature and local directional behavior.

Despite the increased curvature complexity, the mean compatibility error $\bar{E}_{\text{compat}}$ remains equal to zero, indicating that shared-curve synchronization is preserved independently of geometric variation.

4.3 Decor Shelf model

Decor Shelf complements the evaluation with a third non-manifold execution trace whose behavior lies between the two previous cases. It is less uniformly favorable than Book and less sharply diagnostic than Eistute, which makes it useful as an intermediate robustness case on a model with mixed geometric behavior.

Decor Shelf reaches a normalized global error of 0.2605 at Stage 4, corresponding to a total reduction of 73.95% relative to the initial mesh. As shown in Figure 8(a–b), this improvement is weaker than in Book and Eistute, but still substantial. The compact quality indicator $q \geq 0.60$ remains in an intermediate range throughout the trace, oscillating around 50% and recovering to 56.52% in the selected final stage. This does not indicate a dramatic shift toward highly isotropic triangles, but it does show that the method can improve shape quality moderately while still driving down the geometric error.

The most informative aspect of this trace is the recovery pattern. After steady improvement through Stage 2, the execution worsens at Stage 3 and then recovers strongly at Stage 4. The quality distribution in Figure 8(c) shows that this recovery is accompanied by a redistribution away from the lowest-quality range and back toward the intermediate and good-quality bands. In this sense, Decor Shelf supports a restrained but important conclusion: the pipeline is not limited to the best-case behavior observed on Book, and it is able to recover from unfavorable intermediate states without losing the compatibility-preserving character of the rebuilding process.

Decor Shelf also helps reinforce the adaptability argument of the paper. The method is not being evaluated only on a single favorable geometry class. Instead, the reported executions now span a spectrum that includes more structured mixed behavior, strongly curved anisotropic behavior, and intermediate non-manifold surface organization. Decor Shelf occupies this middle ground and shows that, even when quality gains remain moderate, the method can still produce a meaningful reduction of geometric mismatch on a non-manifold model with mixed planar and curved regions.

The results confirm that compatibility is preserved across mixed planar and curved regions, with mean compatibility error $\bar{E}_{\text{compat}} = 0$ for all shared curves.

4.4 Quantitative Summary

The results across all models are summarized in terms of normalized global error (Eq. 17), element-shape quality (Eq. 16), and mean compatibility error $\bar{E}_{\text{compat}}$.

Across the three main evaluation cases, the reported mean compatibility error $\bar{E}_{\text{compat}}$ remains equal to zero, confirming that boundary discretizations are fully consistent across incident patches in the full method.

This result is particularly significant because it shows that, in the full method, compatibility is preserved

across the planar, curved, and mixed geometric settings represented by the main evaluation cases.

4.5 Controlled Comparison with Internal Reference Baselines

Unlike the main evaluation traces discussed above, this controlled comparison uses separate runs designed to isolate the effects of shared-curve synchronization and hybrid reconstruction. These runs are intentionally shorter and were not intended to reproduce the selected final stages reported for the main evaluation cases. They are evaluated at early controlled checkpoints because removing explicit shared-curve synchronization immediately introduces severe boundary cracks and topological inconsistencies. In later stages, these inconsistencies can undermine the robust progression of the advancing-front reconstruction, making the early checkpoints the most stable and interpretable basis for side-by-side comparison with the full pipeline. Within this scope, the controlled runs provide a focused setting for comparing the proposed method against internal baselines that remove one of the two central ingredients of the pipeline. These internal reference baselines should therefore be interpreted as controlled within-pipeline variants rather than as substitutes for broader comparisons against independent external meshing backends.

We performed this controlled comparison on the three reference models: Book, Eistute, and Decor Shelf. The following strategies were evaluated:

- Proposed method;
- Independent-patch adaptive baseline;
- Compatibility-preserving simplified baseline.

The independent-patch adaptive baseline corresponds to the current variant without shared-curve synchronization and represents adaptive rebuilding without explicit cross-patch boundary coupling. The compatibility-preserving simplified baseline corresponds to the current variant without hybrid reconstruction and isolates the effect of preserving boundary coherence without the full interior reconstruction strategy. Table 3 reports controlled checkpoints obtained under this separate comparison protocol; these values are therefore not directly comparable to the stage numbering or element counts reported for the main evaluation traces.

Table 3: Controlled comparison against internal reference baselines. The reported compatibility indicator is the mean compatibility error $\bar{E}_{\text{compat}}$ aggregated over all shared curves and expressed in the native model units of each case.

Model	Variant	Controlled checkpoint	Elements	E_{norm}	$\bar{E}_{\text{compat}}$
Book	Proposed method	1	1064	0.8971	0.00000
Book	Independent-patch baseline	1	556	0.8806	1.34814
Book	Compat.-preserving simplified baseline	1	112	1.0000	0.00000
Eistute	Proposed method	1	80	0.3076	0.00000
Eistute	Independent-patch baseline	1	84	0.3102	5.55846
Eistute	Compat.-preserving simplified baseline	1	64	0.5206	0.00000
Decor Shelf	Proposed method	1	158	0.4475	0.00000
Decor Shelf	Independent-patch baseline	1	140	0.4391	5.28847
Decor Shelf	Compat.-preserving simplified baseline	1	122	0.5837	0.00000

The strongest comparative effect is observed in the mean compatibility indicator $\bar{E}_{\text{compat}}$. Both the proposed method and the compatibility-preserving simplified baseline maintain zero compatibility error in all three models, whereas the independent-patch adaptive baseline produces non-zero incompatibility in every case, with $\bar{E}_{\text{compat}} = 1.34814$ for Book, 5.55846 for Eistute, and 5.28847 for Decor Shelf. These raw displacement values are reported in the native model units and, in the present controlled runs, represent significant macroscopic gaps relative to the local element sizing, visually confirming the loss of watertightness. These resulting boundary cracks typically manifest as overlapping elements or open T-junctions that directly prevent the valid closure of adjacent fronts. This comparison shows that the

decisive difference is not adaptive rebuilding alone, but whether neighboring patches are forced to share the same updated boundary discretization.

The geometric effect is more nuanced. The independent-patch adaptive baseline does not uniformly worsen the selected normalized global error in the present controlled runs, which indicates that explicit synchronization should be interpreted primarily as a structural compatibility mechanism rather than as a direct geometric-error reduction device. For instance, on the Book model, the independent baseline achieves a slightly lower normalized error using roughly half the elements (556 vs. 1064). This highlights the intrinsic cost of non-manifold compatibility: enforcing a synchronized parameter sequence along a multi-incident curve inherently forces the insertion of additional elements and imposes stronger geometric constraints on the interior reconstruction. However, this lower element count is obtained at the cost of a severe topological mismatch, with $\bar{E}_{\text{compat}} = 1.34814$ instead of zero. For Eistute, the independent-patch baseline slightly changes both the element count and the normalized error at the reported early checkpoint (84 and 0.3102 versus 80 and 0.3076), while for Decor Shelf the same desynchronization changes them more visibly (140 and 0.4391 versus 158 and 0.4475). In both cases, the geometric values remain of the same order, yet the compatibility indicator deteriorates sharply. This reiterates that unconstrained boundaries trade topological soundness for comparatively minor local geometric variations at these early checkpoints. This behavior is consistent with runs in which boundary desynchronization appears before larger interior divergence accumulates.

By contrast, the compatibility-preserving simplified baseline consistently weakens geometric adaptation relative to the proposed method. The normalized error worsens from 0.3076 to 0.5206 in Eistute, from 0.4475 to 0.5837 in Decor Shelf, and from 0.8971 to 1.0000 in Book. In the case of Book, the simplified variant essentially stalls at the level of the initial geometric approximation. This shows that enforcing strict boundary synchronization introduces strong geometric constraints that cannot be converted into meaningful geometric improvement by a static interior alone; the hybrid interior reconstruction is therefore not merely a quality enhancer, but a necessary component for accommodating the synchronized shared-curve samples while improving the geometric fit.

Taken together, this controlled comparison supports the interpretation that shared-curve synchronization is the central mechanism for compatibility preservation, while hybrid reconstruction is the central mechanism for stronger geometric improvement.

4.6 Discussion of Results

The results consistently show that compatibility preservation is independent of geometric complexity. While normalized global error (Eq. 17) captures geometric fidelity and the element-shape metric (Eq. 16) reflects mesh regularity, the mean compatibility error provides complementary information by explicitly measuring boundary coherence.

Taken together, the three reported evaluation cases support a focused but coherent interpretation of the proposed method.

First, they support the central compatibility-preserving claim of the paper: adaptive rebuilding can be carried out on non-manifold parametric patch complexes while maintaining coherent shared-curve behavior across incident patches and substantially reducing the global geometric error indicator. Second, they show that the shape response of the mesh is model-dependent rather than uniformly monotone. This is not a contradiction of the method, but part of the correct interpretation of its current scope.

Book provides the clearest positive case: its selected final stage combines a normalized global error of 0.0843, corresponding to a 91.57% reduction relative to the initial mesh, with a strong shift toward the highest quality band and a very high share of elements with $q \geq 0.60$. Eistute shows the strongest geometric adaptation: its selected final stage reaches a normalized error of 0.0367, corresponding to a 96.33% reduction, but with only moderate gains in isotropic 3D shape quality. Decor Shelf occupies an intermediate position, combining substantial error reduction with moderate but non-negligible improvement in shape distribution.

The results therefore support a qualified but meaningful conclusion. The proposed pipeline already provides evidence that compatibility-preserving adaptive rebuilding is feasible and effective on non-manifold

patch complexes with different geometric profiles, including models with more structured regions, mixed planar/curved organization, and stronger local curvature and anisotropy. At the same time, the present evaluation also shows why the principal metric for this class of problem should remain the geometric surface error, with isotropic triangle shape indicators interpreted as complementary descriptors rather than as universal final judges of mesh quality. This boundary is important: it keeps the interpretation of the results aligned with what the present manuscript actually demonstrates.

Most importantly, the controlled comparison against internal reference baselines demonstrates that shared-curve synchronization is the central mechanism responsible for boundary consistency, while hybrid reconstruction is the main contributor to stronger geometric improvement, confirming the main hypothesis of this work.

4.7 Runtime Analysis

Table 4: Average runtime over 10 runs per model in sequential evaluation mode for the proposed method.

Model	Wall time (s)	Non-reconstruction measured work (s)	Patch reconstruction (s)
Book	148.50	20.89	127.61
Eistute	13.20	1.77	11.43
Decor Shelf	7.44	0.43	7.01

Runtime measurements were obtained on a Dell OptiPlex 3070 equipped with an Intel Pentium Gold G5420T CPU at 3.20 GHz (2 physical cores, 4 logical threads) and 15.81 GB of RAM, running Windows 11 Home 64-bit. The code was compiled in Release mode with CMake using MinGW Makefiles and optimization flags `-O3 -DNDEBUG`. All experiments reported in this section were executed in sequential evaluation mode with 1 process and 1 thread, and the reported timings were averaged over 10 runs for each model.

The partial quantitative runtime analysis reported in Table 4 confirms that patch-interior reconstruction dominates the computational cost of the proposed pipeline in all three evaluation cases. For Book, patch reconstruction accounts for 127.61 s out of a total wall time of 148.50 s, leaving 20.89 s for the remaining measured work summarized in the non-reconstruction measured work column. For Eistute, patch reconstruction accounts for 11.43 s out of 13.20 s, leaving 1.77 s. For Decor Shelf, patch reconstruction accounts for 7.01 s out of 7.44 s, leaving 0.43 s. Even under this aggregated reading, the non-reconstruction cost remains secondary relative to patch-interior reconstruction.

The larger wall time observed for Book is also consistent with its patch-complex composition: unlike Eistute, which contains only 4 strongly curved patches, and Decor Shelf, which contains 13 patches with only one curved patch, Book contains 28 curved patches. Since the dominant cost of the present pipeline lies in repeated patch-interior reconstruction, this substantially larger number of curved patch domains leads to a much higher cumulative reconstruction cost, even when the final element count is not dramatically larger than in the other models.

This quantitative behavior is consistent with the design of the method. Compatibility preservation is enforced explicitly throughout the adaptive process, but the dominant cost remains the adaptive rebuilding of full patch interiors, which requires repeated quadtree refinement, front propagation, local feasibility checks, and mesh improvement over entire patch domains. The numeric evidence therefore supports the claim that the reconstruction stage, rather than the auxiliary work grouped outside it, is the runtime bottleneck of the pipeline.

The present runtime evaluation should be interpreted as a partial quantitative runtime analysis rather than as a complete stage-by-stage performance benchmark. At the current stage of instrumentation, wall time, shared-curve update, and patch reconstruction are the components that can be reported reliably, whereas a finer decomposition of all internal stages is not yet stable enough for publication. As a consequence, the runtime evidence is most useful for identifying the dominant stages of the workflow and for confirming that compatibility enforcement is not the main bottleneck. More complete stage-level profiling, broader performance evaluation, comparison against additional implementations, and parallel scaling are left for

future work.

Overall, the runtime profile indicates that the proposed compatibility mechanism remains computationally secondary relative to the dominant cost of adaptive patch reconstruction.

5 Discussion, Limitations, and Future Work

5.1 Discussion

The proposed pipeline should be understood as a practical adaptive meshing method for non-manifold parametric patch complexes, with explicit shared-curve compatibility treated as a structural constraint throughout the rebuilding process. Within this scope, the current results support the central claim of the paper: adaptive rebuilding can be combined with shared-curve synchronization so that mesh refinement remains compatible across incident patches, including multi-incident configurations, while still achieving meaningful reduction of the global geometric error indicator.

This contribution is best interpreted as a compatibility-centered meshing strategy rather than as a replacement for all existing surface-meshing paradigms. The method is not introduced as a fully general anisotropic framework, nor as a quality-guaranteed meshing kernel. Instead, its contribution lies in the interaction between three ingredients that are usually treated separately: curvature-guided adaptation, patch-wise rebuilding in parameter space, and explicit synchronization along shared curves. The experiments reported in section 4 are consistent with this interpretation. In particular, Book provides the clearest all-around positive case, combining strong geometric-error reduction with strong isotropic shape improvement, whereas Eistute shows that on more curved and locally anisotropic models, geometric adaptation can be substantially stronger than what is suggested by a purely isotropic 3D shape indicator. Decor Shelf, in turn, reinforces the adaptability of the method on a mixed geometric setting, with meaningful error reduction and moderate but non-negligible gains in element-shape quality.

A second important point is that the current implementation adopts a deliberately pragmatic design. The factor-driven refinement policy is simpler than a full metric-tensor formulation, but that simplicity is part of what allows the method to remain tightly coupled to the shared-curve policy. Likewise, the hybrid interior-generation stage combines structural support and local flexibility without claiming the formal guarantees associated with restricted Delaunay or Frontal-Delaunay formulations. This engineering balance is appropriate for the present contribution, whose main objective is to demonstrate that adaptive rebuilding and non-manifold compatibility can be integrated coherently in a single pipeline.

The strongest current evidence concerns compatibility preservation. Across the reported evaluation cases, the mean compatibility error $\bar{E}_{\mathrm{compat}}$ remains equal to zero, and the controlled comparison against internal reference baselines shows that removing shared-curve synchronization immediately introduces measurable boundary incompatibilities. At the same time, the geometric effect of synchronization is less direct, whereas the removal of hybrid reconstruction produces a clearer degradation in normalized global error. The present results therefore support the interpretation that shared-curve synchronization is the principal mechanism for preserving boundary coherence, while hybrid reconstruction is the principal mechanism for stronger geometric improvement.

For this reason, the present manuscript should be read as establishing a clear methodological direction and an experimentally supported first contribution, rather than as exhausting the space of adaptive meshing strategies for non-manifold geometry. The strongest evidence currently provided is that compatibility along shared curves can be maintained as a first-class adaptive constraint without giving up meaningful geometric adaptation, even across models with different mixtures of planar, mildly curved, and strongly curved regions. The next step is therefore not to redefine the problem, but to consolidate the empirical and comparative validation of the current design.

5.2 Limitations

Despite its robustness, the proposed method has several limitations. First, the current refinement strategy is primarily isotropic in the parametric domain and does not explicitly account for anisotropic adaptation aligned

with principal curvature directions. This design reflects a deliberate choice to prioritize a compatibility-preserving reconstruction workflow that remains tightly coupled to shared-curve synchronization. As a result, the present method emphasizes robust boundary coherence before pursuing a more general anisotropic formulation, which remains a natural extension for future work.

Second, the method operates under the assumption that the underlying patch parametrization is well-defined and sufficiently regular. In cases of highly distorted parametrizations, the quality of the resulting mesh may be affected.

Third, while the compatibility mechanism ensures consistent boundary discretization, it may introduce additional constraints that restrict local refinement flexibility near shared curves, potentially leading to suboptimal element distributions in certain configurations.

Fourth, the current evaluation is still limited in scope. Although the evaluated cases cover planar, curved, and mixed geometric settings, the present study is intentionally focused on controlled non-manifold patch-complex cases chosen to isolate compatibility-preserving rebuilding behavior. The current manuscript already includes controlled comparison against internal reference baselines, but this evidence should be interpreted as controlled within-pipeline validation rather than as a definitive external benchmark study. Broader validation against independent meshing backends and larger benchmark suites therefore remains an important next step for establishing how the proposed design generalizes across a wider range of geometric and topological conditions.

5.3 Future Work

Several directions can be explored to extend the proposed framework. A natural extension is the incorporation of anisotropic refinement strategies aligned with local curvature directions, which could improve efficiency and element quality in strongly directional surface regions.

A second important direction is a broader experimental validation, beginning with comparisons against external meshing backends and larger benchmark sets. Such experiments would extend the controlled internal reference-baseline analysis reported here and help clarify how the present compatibility-preserving design compares with other feature-aware and adaptive meshing strategies under more diverse geometric and topological conditions.

Another promising avenue is the refinement of the compatibility analysis itself, including more detailed local indicators for shared-curve consistency and junction behavior in highly complex multi-incident configurations.

Finally, the current implementation could be extended toward a more comprehensive runtime study and a more reproducible experimental package, including standardized execution settings, additional logs, and public benchmark artifacts.

6 Conclusion

We presented a compatibility-preserving adaptive meshing pipeline for non-manifold parametric patch complexes with shared multi-incident curves. The method combines curvature-guided adaptation, shared-curve synchronization, and hybrid patch-interior reconstruction within a single evaluate-and-rebuild workflow.

The reported evaluation cases show that the proposed pipeline can substantially reduce normalized global geometric error while preserving coherent boundary discretization across incident patches. Across the evaluated cases, this behavior is observed in planar, curved, and mixed geometric settings, supporting the applicability of the method to a representative range of non-manifold configurations.

The experimental analysis also clarifies the internal role of the main components of the pipeline. The controlled comparison against internal reference baselines indicates that shared-curve synchronization is the central mechanism for preserving compatibility, whereas hybrid reconstruction is the central mechanism for stronger geometric improvement. Together, these results support the main methodological claim of the paper: compatibility can be treated as a first-class adaptive constraint rather than as a post-processing correction.

At the same time, the present work should be understood as a methodologically specific contribution within a broader design space. The current implementation is primarily isotropic, relies on a practical factor-driven adaptation policy, and has been evaluated on a limited but representative set of cases. Future work should therefore expand the comparative validation against independent external backends, improve anisotropic adaptation, and further strengthen reproducibility and runtime analysis.

Under this interpretation, the main contribution of the paper is not a universal meshing solution, but a compatibility-centered adaptive pipeline for non-manifold parametric patch complexes in which shared-curve coherence must be preserved throughout rebuilding.

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